

PRE-WORKOUT OPTIMIZATION GUIDE

Ch 1: Key Ingredients Science

Understanding the science behind each pre-workout ingredient allows you to build a personalised, cost-effective stack that outperforms most commercial products. Many commercial pre-workouts underdose key ingredients — knowing effective doses is essential.

Ingredient	Effective Dose	Mechanism	Benefit	Timing
Caffeine	150–300 mg	Adenosine antagonist; CNS stimulant	Energy, focus, fat oxidation, endurance	30–45 min pre-workout
L-Theanine	200–400 mg (2:1 with caffeine)	Alpha-wave promotion; reduces overstimulation	Smooth energy; no jitters; no crash	Same time as caffeine
Citrulline Malate	6–8g (2:1 ratio)	NO precursor; arginine sparing	Pumps, endurance, reduced DOMS	30–45 min pre-workout
Beta-Alanine	3.2–6.4g/day (split doses)	Carnosine precursor; buffers H ⁺ ions	Delays muscular fatigue; endurance	Daily loading; not just pre-workout
Creatine Monohydrate	5g/day	ATP-PCr system replenishment	Strength, power, muscle fullness	Any time; consistency is key
Betaine Anhydrous	2.5g	Osmolyte function similar to creatine	Power output; hydration; body comp	Pre-workout with meal
L-Tyrosine	500–2000 mg	Catecholamine precursor (dopamine, adrenaline)	Focus under stress; mental performance	30 min pre; best fasted
Nitrosigine	1.5g	Enhanced arginine silicate complex	Pumps superior to L-arginine	30 min pre-workout

Ch 2: Pre-Workout Stacks by Goal

Goal	Ingredients & Doses	DIY Cost/Month	Commercial Equivalent
Clean energy + focus	Caffeine 200mg + L-Theanine 200mg + L-Tyrosine 1g	Rs 400–600	Most basic pre-workouts; look for 200mg caffeine + theanine
Maximum muscle pump	Citrulline malate 8g + Nitrosigine 1.5g + Beetroot extract 500mg	Rs 700–1000	Stimulant-free pump products; ideal for evening training
Endurance enhancement	Caffeine 150mg + Beta-Alanine 3.2g + Creatine 5g + Sodium 500mg	Rs 500–700	Endurance pre-workouts; often labelled for cardio athletes
Maximum strength output	Creatine 5g + Betaine 2.5g + Caffeine 200mg + L-Theanine 200mg	Rs 600–900	Strength-focused pre-workouts; check for betaine and creatine
Complete DIY pre-workout	Caffeine 200mg + Theanine 200mg + Citrulline 8g + Beta-Alanine 3.2g + Creatine 5g	Rs 900–1400	Outperforms most commercial products at a fraction of the cost

Ch 3: Caffeine Optimization Protocol

Getting Maximum Benefit from Caffeine

- Delay first coffee/tea to 90 minutes after waking — let the natural cortisol peak fade first
- Cycle caffeine: 5 days on, 2 days off to prevent tolerance build-up
- Always pair with L-Theanine (2:1 ratio Theanine:Caffeine) for clean, focused energy
- Last caffeine dose by 2 PM to protect sleep (caffeine half-life is 5–6 hours)
- Total daily maximum: 400mg for healthy adults (approximately 3–4 cups strong coffee)
- Sources: coffee, black tea, green tea, caffeine anhydrous powder or pills — all equivalent
- Avoid: proprietary blend products hiding caffeine dose; energy drinks with added sugar

Pre-Workout Product Check	Pass	Fail
Caffeine dose listed clearly?	200mg+ listed on label	Proprietary blend; dose hidden
Citrulline dosed at 6–8g?	6g+ citrulline malate listed	Under 3g or arginine used instead
Beta-alanine at 3.2g+?	3.2g+ per serving	Under 1g (tingling with no benefit)
No banned substances?	Transparent ingredient list	DMHA, DMAA, ephedrine present

Ch 4: Indian DIY Pre-Workout Recipes

Recipe	Ingredients	Cost Per Serving	Best For
Basic Desi PWO	Strong black coffee + 5g creatine + pinch salt	Rs 15–20	Everyday training; minimal budget
Pump PWO (no stim)	Watermelon juice 300mL + citrulline 6g + creatine 5g	Rs 25–35	Evening training; already caffeinated
Advanced DIY PWO	2 strong green teas + L-theanine + beta-alanine 3.2g + creatine	Rs 20–30	Clean energy; good focus; budget-friendly
Natural whole-food PWO	Banana + strong chai (no sugar) + 100g roasted beetroot	Rs 30–40	No supplements needed; good for beginners
Ultra Budget PWO	Instant coffee 2 teaspoons + 5g creatine powder in 500mL water	Rs 10–15	Absolute minimum; still effective

Ch 5: Timing, Tolerance & Safety

Scenario	Pre-Workout Recommendation	Notes
Morning training (6–8 AM)	Full pre-workout stack 30 min before	Natural cortisol peak supports the effect
Lunchtime training (12–2 PM)	Moderate caffeine 150mg; full citrulline + creatine	Cortisol falling; caffeine helps sustain focus
Evening training (6–8 PM)	Stimulant-free pump stack OR half-dose caffeine	Caffeine after 5 PM disrupts sleep; use citrulline + beta-alanine
Night training (8–10 PM)	Stimulant-free only: citrulline + creatine + beetroot	No caffeine; focus on pump and performance without stims
Fasted training AM	Include carbs 30 min before if session > 45 min	Without carbs, high-intensity training quality drops

Pre-Workout Supplement Safety Rules

- Always start with half the serving and assess tolerance before full dose
- Never mix multiple stimulant products (double caffeine dose is a cardiac risk)
- Do not use pre-workouts on rest days — avoid building tolerance without training benefit
- People with heart conditions, high blood pressure, or anxiety: avoid stimulant pre-workouts
- If you feel chest tightness, irregular heartbeat, or extreme anxiety: stop immediately